Introduction

Airbnb is quickly changing the travel industry with its multi-dimensional characteristics including features like price flexibility, various location choices, and reviews from guests and hosts. With low entry barriers and flexible business structure as a means of extra income, more people have been motivated to list their properties. Under this circumstance, the understanding of determinants of the total revenue has become significantly important for hosts and investors to make business decisions.

There have been some studies on the performance of Airbnb. Wang and Nicolau (2017) investigated price strategies and found that listing price are mainly influenced by room type and location. Kwok and Xie (2019) also found that multi-unit hosts could have higher revenues compared with single-unit hosts, because they gained more experiences on price setting. However, revenue was not only influenced by price but also other explanatory variables. Gunter and Önder (2018) studied the elasticity of Airbnb demand and found that the revenue of hosts depended on the host responsiveness and listing capacity.

Our study integrates these studies and overcomes their limitations in the scope. We aim to study the relationship between total revenue and a set of explanatory variables including host experience, nightly price, customer review scores, listing capacity, host responsiveness, and room type. We expect that the listings with higher price, better scores, more capacity, more private personal space and faster responsiveness would have higher total revenue. To model the relationship between total revenue and explanatory variables, we use multiple linear regression approach. MLR is suitable for our study as we want to study the marginal effect of each explanatory variable on total revenue while holding others constant. We aim to maintain model interpretability by applying only one transformation per variable. However, our objective is to have more precise predictions. With this model, hosts could input fixed listing characteristics and see how the expected total revenue would change if it responded to changes in more controllable variables.

Understanding the Airbnb revenue would be useful for hosts who would like to make more earnings with their decisions on price, quality of service, and the amenities of the listing. It would also be useful for investors who would like to rent their properties to Airbnb.

Data Description

Rubric:

Description of data source:

- Where the data was sourced/downloaded from is explicitly mentioned with a corresponding citation in the bibliography.
- The original usage or purpose of the dataset is described, and it is explicit how that usage differs from the current research proposal.
- How the curator of the dataset originally collected the data is described and a corresponding reference is cited from the bibliography.

Response variable summary:

- An appropriate and suitably presentable numerical or graphical summary is used to describe the response variable statistically.

- A written description of the response variable highlights important features of the response distribution, in the context of what is being measured/the research question.
- A justification for why the chosen response variable is suitable to be used in a linear regression model is provided and is correct, based on the statistical summary presented

Predictor variable summaries:

- An appropriate and suitably presentable numerical or graphical summary is used to statistically describe the chosen predictor variables.

- Important/interesting variable characteristics (e.g. skews, abnormal values) or lack thereof are, in the context of what is being measured/the research question
- A justification for why the chosen predictor variables are relevant to answering the research question, making explicit reference to any modifications to variables that have been made

To investigate the total revenue of an Airbnb property, we used the dataset on Airbnb properties in the Netherlands (Cannata, 2017). The details suggested that the data was likely sourced from Airbnb backend logs to predict customer ratings for Airbnb properties. It was sufficient for our purpose of predicting the total revenue of a property to advise optimal business strategies.

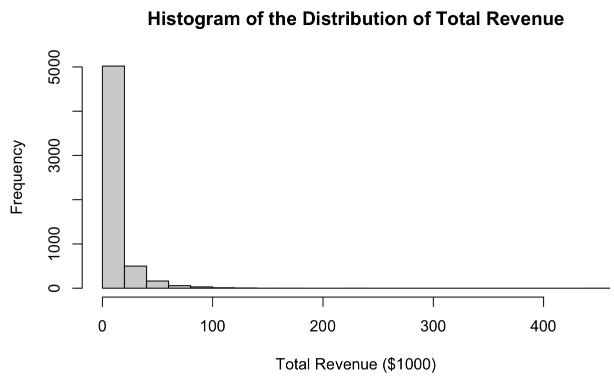
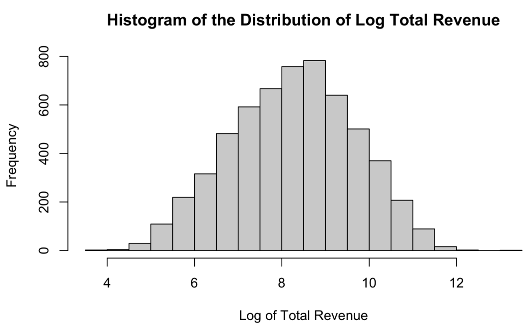
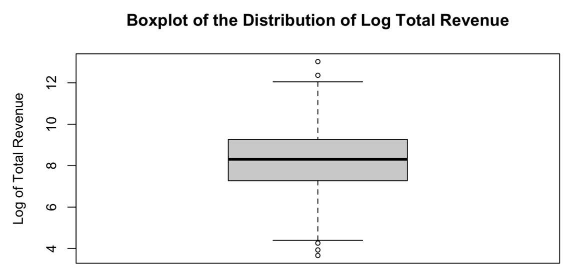


Figure 2.1: Summary of total revenue.



(a)



(b)

Figure 2.2: Summary of the log of total revenue.

The total revenue distribution was right-skewed, mostly below $100,000 (Figure 1). Since total revenue was positive, a logarithmic transformation was able to remove its skewness. Log total revenue appeared roughly normal(Figure 2(a)), exhibiting very few outliers (Figure 2(b)). As a continuous variable, the log of total revenue showed its potential as a response for our linear regression model.

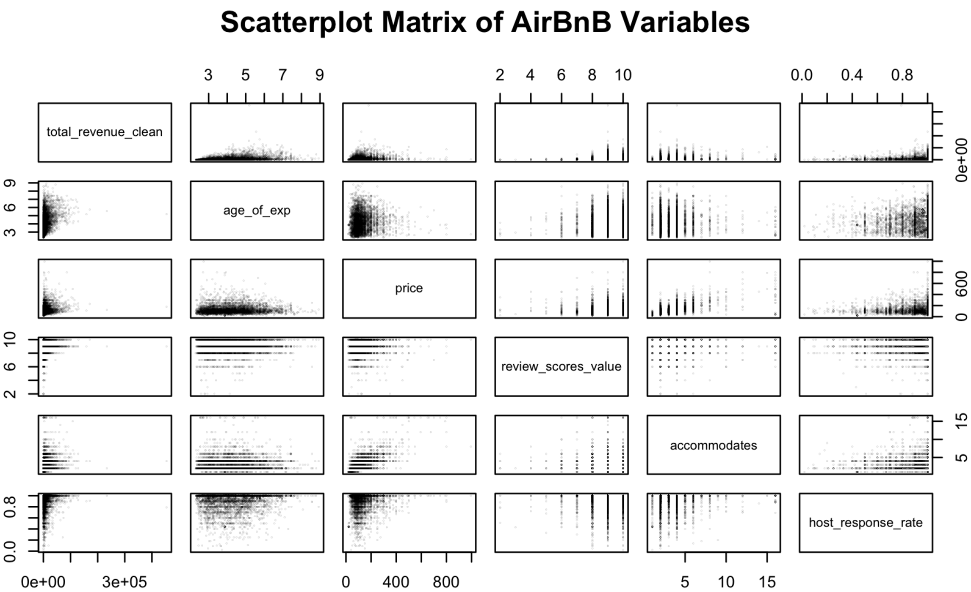


Figure 2.3: Analysis of relations between candidate variables.

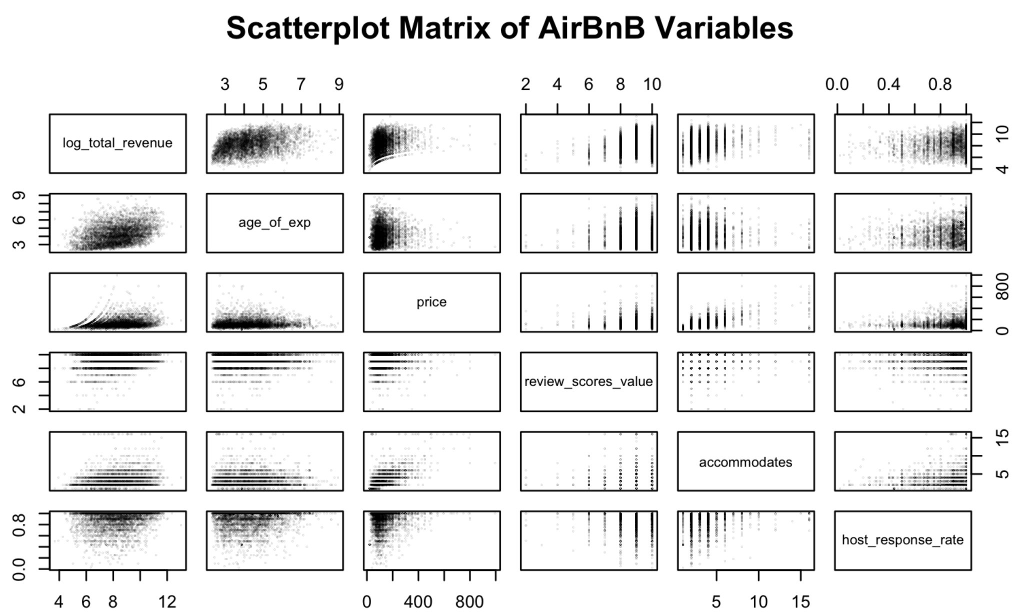


Figure 2.4: Analysis of relations between preliminary predictors and the log of total revenue.

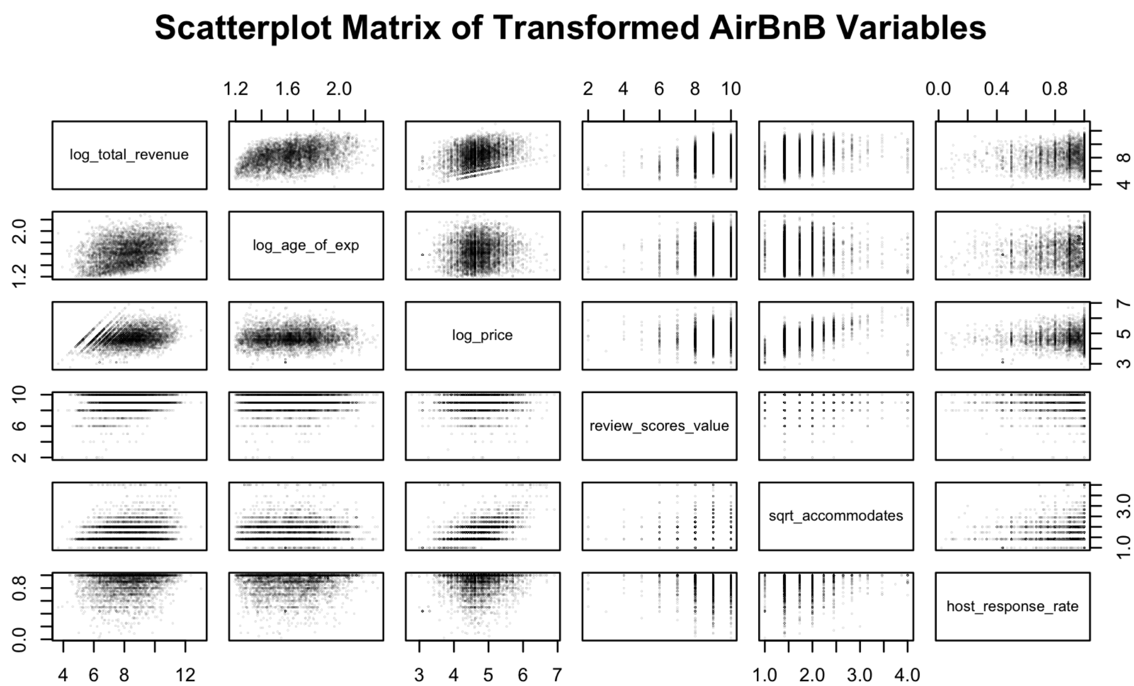


Figure 2.5: Analysis of relations between transformed preliminary predictors and the log of total revenue.

We chose five continuous variables that potentially impact total revenue–age of experience of host, review scores value, accommodates and host response rate–and verified their interactions via a scatterplot (Figure 2.3). When checking their relations with log total revenue, both review scores value and host response rate exhibited linearity (Figure 2.4). Age of experience and price against log total revenue plots were right-skewed and were linearized by a logarithmic transformation (Figures 2.4, 2.5). The accommodates versus log total revenue plot showed a parabolic shape, and a square root transformation linearized it (Figures 2.4, 2.5). Since all variables mentioned are positive, the above transformations were legal.

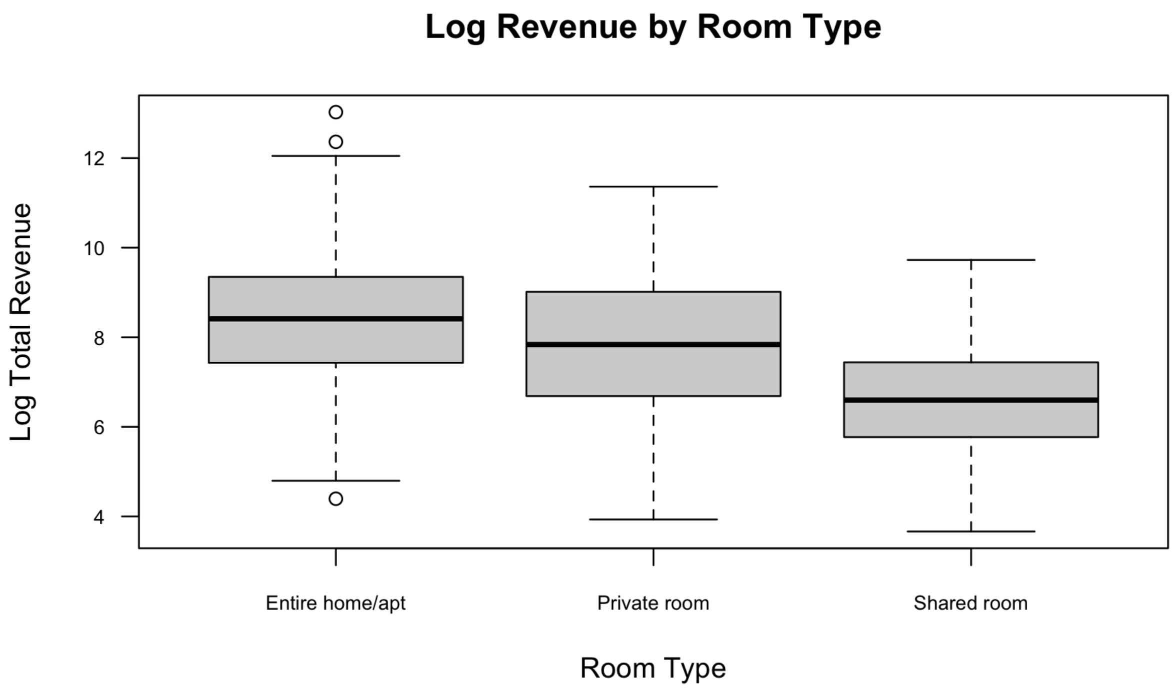
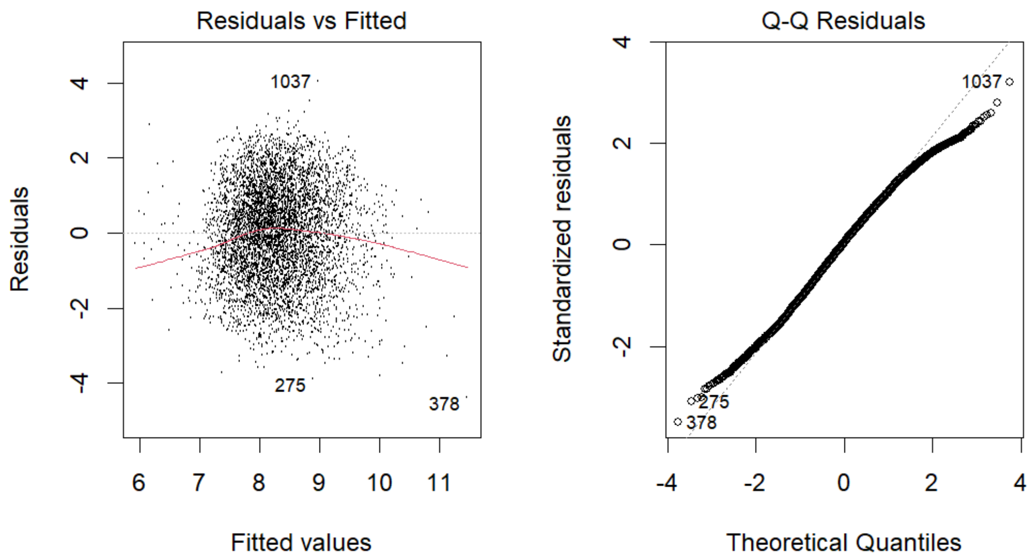


Figure 2.6: Analysis of relations between log revenue and room type.

Analysis of another potential factor, room types, showed that the log total revenue of entire homes/apartments centred at the highest value amongst the three room types, followed by private rooms and shared rooms (Figure 2.6). Even with slight variations in IQR and outliers, the distributions for all categories were reasonably similar with similar spreads(Figure 2.6). This verified the relationship between room type and log total revenue. Hence, room type was a reasonable categorical predictor.

Preliminary results

Residual analysis of preliminary model:



First and foremost, we create the scatter plot with residuals against corresponding fitted values. According to regression diagnostics, when the model is correct, the residuals are uncorrelated with the fitted value and predictors.

In this graph, we can see there is no obvious correlation between them and the fitted values. Furthermore, this is approximately a null plot with no systematic pattern, which can demonstrate the fact that a scatter plot of residuals against fitted values should have a constant mean of 0.

Secondly, we have a QQ plot checking the normality of standardized residuals. We can observe a N(0,1)-ish distribution since our set of residuals fits the QQ plot of a normal distribution. Therefore, this diagram provides plausible evidence that our model passes the check.

Linear assumption

- Residual against fitted values → null plot
- Response against predictor → linear approx

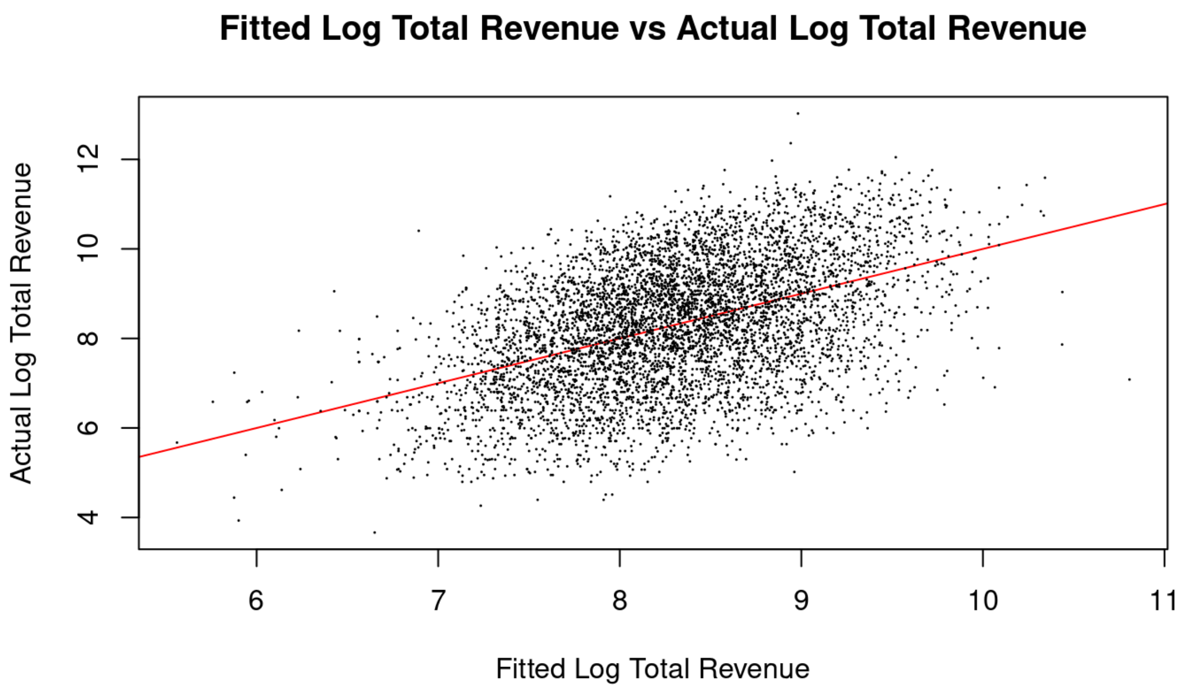
Homoscedasaiticity

- Residual should have constant variance → residual plot vary without obvious patterns around 0

Normality of error

- QQ plot

Preliminary model discussion:



Interpret each estimated coefficient from beta0 to beta7:

1. For β0 hat, it is the intercept of our model and it represents the initial value of the hotel's revenue when disregarding the categorical variable room_type.
2. For β1 hat, when log_age_of_exp increases by 1%, the log_total_revenue increases by 2.29809%.
3. Similar to β1, β2 to β5 are slopes of the model related to different numeric predictors (from age of exp to host response rate). A 1% change in the predictor variable will cause a corresponding βn% change in our response variable.
4. For β6 and β7, they take effect only when the dummy variable takes 1. For example when the room is private, then β6+β0 will be the new intercept.

Taken together, we believe the estimates suggest that both host experience (a numerical factor) and listing type (a categorical factor) play critical roles in determining revenue outcomes. For hosts or platforms aiming to improve performance, encouraging long-term host engagement and promoting more private or exclusive listings may be effective strategies.

Tab 6

To check the validity of our linearity assumption, we will be using Residual vs Fitted Scatterplot, and Standardized Residual Histogram, QQ plot.

Despite exhibiting a slight U-shape, the Residual vs Fitted model is nearly 0-mean throughout and homoskedastic (Figure 3.1(a)), meaning that it is close to a null plot; this demonstrates minimal correlation between the residual and the fitted values. Both characteristics suggest that the linear assumption is valid.

Although our QQ-plot suggests that our model might not exactly have a $N(0,1)$ standardized residual with heavy tails (Figure 3.1(b)), the distribution look approximately like $N(0,1)$, as required by linear assumption (Figure 3.1(c)).

Therefore, our residual analysis shows that a linear relationship is still plausible, with further room for explainability present in our data.

In conclusion, our preliminary fitted model is given by:

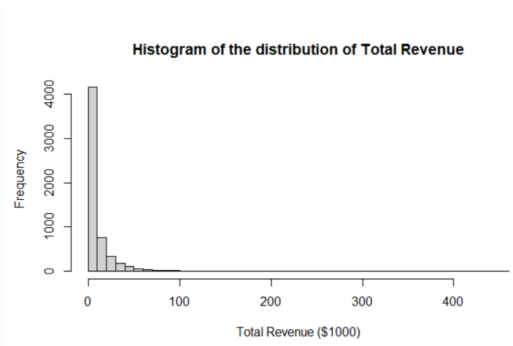
$$
\begin{align*}
\log(\text{Total Revenue}) =& -1.80055 + 2.31110\cdot \log(\text{Years of Experience}) + 0.48899\cdot \log(\text{Price})\\
&+ 0.10537\cdot (\text{Review Scores Value}) + 0.32137 \cdot \sqrt{\text{Accommodates}} \\
&+ 0.94504\cdot (\text{Host Response Rate}) - 0.07089\cdot (\text{is Private Room}) \\
&- 0.85165\cdot(\text{is Shared Room}).
\end{align*}
$$

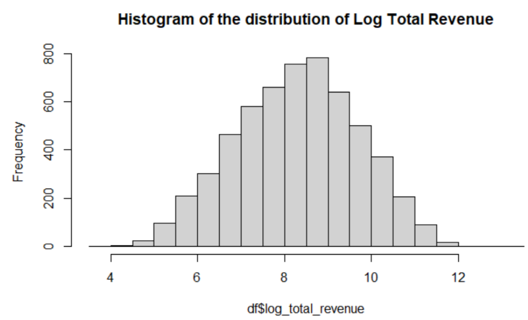
This demonstrates that any increase in a continuous predictor variable while holding all other predictors constant, such as Years of Experience as a host, would lead to an increase total revenue. In particular we see that increasing $\log(\text{Price})$ by 1 would increase $\log(\text{Total Revenue})$ by 0.48899 (all else constant), which demonstrates that AirBnB is demand-inelastic, which agrees with results found by Wang and Nicolau (2017).

Moreover, by our categorical variable Room Type, we see that leasing out the entire home, private room, and shared room generates revenue in decreasing order. This is expected as entire home and private rooms give more privacy, translating to better homestay experience, thus leading to higher revenue generated.

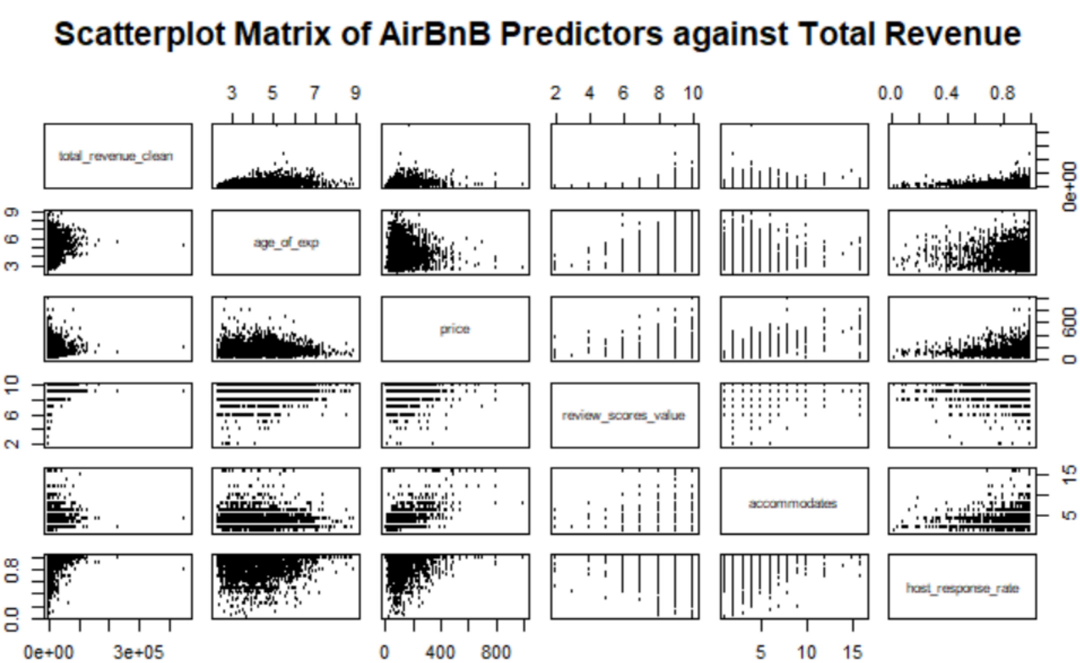
Model Selection

In fitting our multiple linear regression model, we carefully examined each continuous and categorical predictor to ensure the model meets all assumptions of linear regression and is correctly specified. To achieve this, we introduced several transformations of the predictors, each of which can be justified based on the data and model requirements.

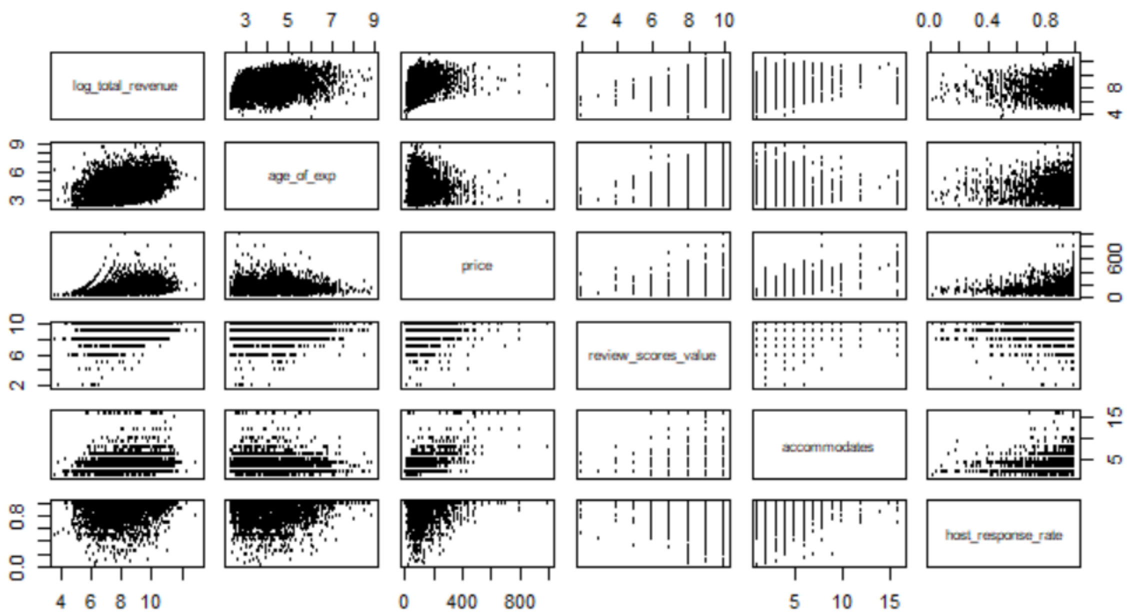




The first transformation we apply to our model is log(y), where y is the total revenue (the response variable). Although the normality of Y is not an assumption of MLR, we still find advantages in doing this. For instance, if Y is highly skewed, it's often a sign that residuals might also be skewed, violating an important regression assumption. In such cases, transforming Y (e.g., log(Y)) can normalize the distribution of residuals and even Improve model fit and inference. For more accuracy, its positive effect also can be seen in increasing linearity between y and other predictors. See the following scatterplot matrices:

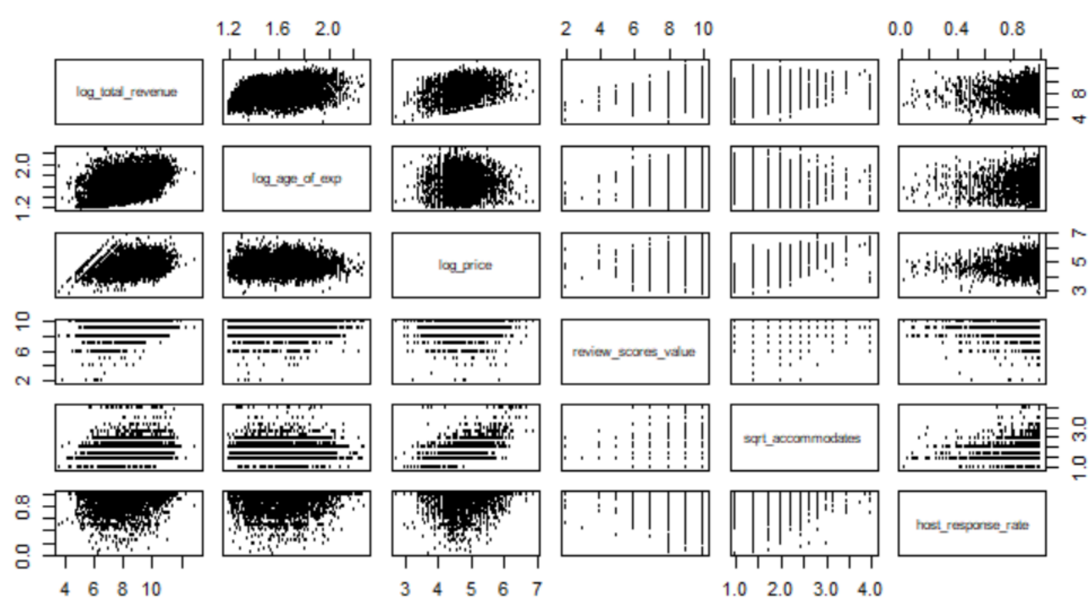


Scatterplot Matrix of AirBnB Variables against Log Total Revenue



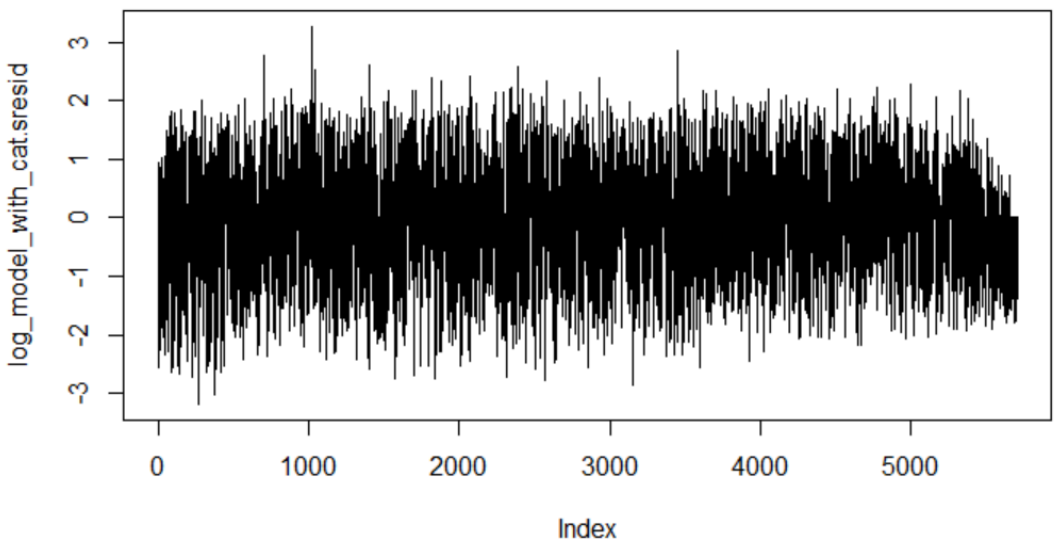
As we can observe, especially for scatterplots of response variables vs continuous predictors, the trend is more visible and offers a great initiation of fitting multivariate regression. Besides, we have also transformed some chosen predictors that contribute to better quasi-linear behaviour, such as adding powers to terms.

Scatterplot Matrix of Transformed Predictors against Log Total Revenue

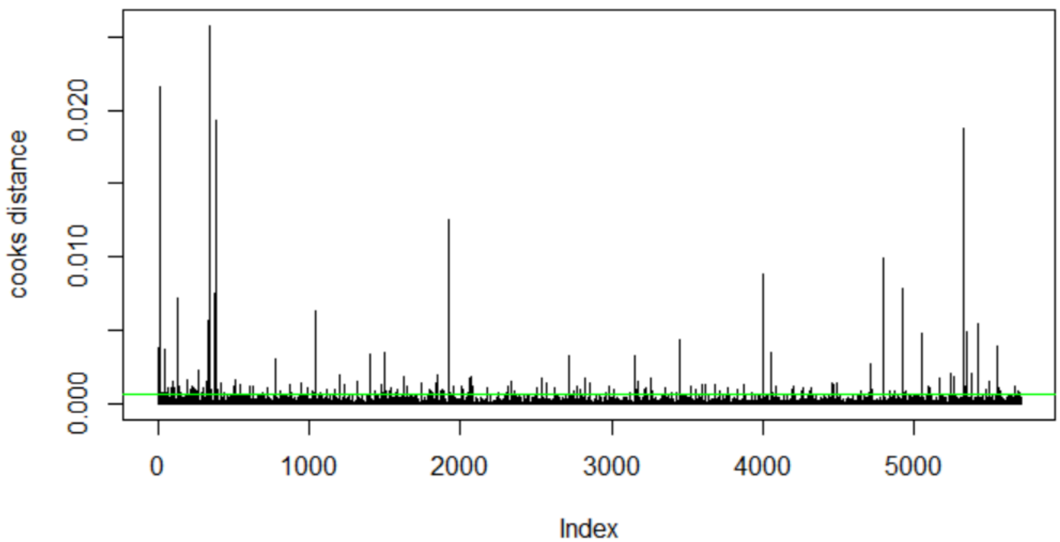


To further explore the potential of transformations, we tested the box-cox transformation. We expect that it can stabilize variance and improve model fit for regression. However, it turns out that this method does not give us significant improvements and even makes interpretation much more difficult. Taking these factors into account, we finally decided not to carry on doing this in our model.

For outliers, we found out that if using 4 as the criterion to assess the standardized residuals, there will be no outliers in this cleaned dataset. We did not use 2 as the criterion since for such a large sample, 2 will be too strict and sensitive. It is more understandable to think intuitively: If residuals are normally distributed, then ~5% of them will naturally fall outside ±2, and ~0.3% will fall outside ±3. Therefore, with 10,000 observations ~500 might exceed ±2 just due to random variation and ~30 might exceed ±3. We cannot claim all 30 are outliers because they are reasonable to exist.



For influential points, we use the cook’s distance as an indicator, and the green horizontal line at y = (4 / number of rows in the dataset) as the criterion to look for influential points. According to this standard, multiple influential points are detected and they have relatively much more potential than normal data points. Upon review, most of them correspond to usual cases where the hotel's revenue is much higher or lower than competitors due to many practical reasons. In light of the large number of influential points and other factors in reality, we determined not to remove them in order to keep unbiasedness. What is worse about excluding influential points is that it could lead to the loss of important information or biased parameter estimates.



After handling the problematic observations and influential points, we proceed to check our selection of predictors. The first step we take is to inspect the necessity for adding an extra categorical predictor room_type to the model. By conducting a partial F test on the reduced model (without room_type) with the full model (with room_type), we witnessed a slightly lower RSS, meaning it can better explain the data. Furthermore, we gain a higher F that demonstrates effectivity brought by adding room_type. Our introduction of this new variable is statistically significant since the p-value is much smaller than 0.5.

```
Analysis of Variance Table

Model 1: log_total_revenue ~ log_age_of_exp + log_price + review_scores_value +
    sqrt_accommodates + host_response_rate
Model 2: log_total_revenue ~ log_age_of_exp + log_price + review_scores_value +
    sqrt_accommodates + host_response_rate + room_type
  Res.Df    RSS Df Sum of Sq      F   Pr(>F)    
1   5696 8744.5                                 
2   5694 8718.2  2    26.294 8.5866 0.000189 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Out of curiosity we gave another categorical variable a try and wanted to see the result of the ANOVA test. The name of the new variable is host_response_time.

```
Analysis of Variance Table

Model 1: log_total_revenue ~ log_age_of_exp + log_price + review_scores_value +
    sqrt_accommodates + host_response_rate + room_type
Model 2: log_total_revenue ~ log_age_of_exp + log_price + review_scores_value +
    sqrt_accommodates + host_response_rate + room_type + host_response_time
  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
1   5694 8718.2
2   5691 8635.4  3    82.806 18.191 9.605e-12 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

The outcome is that we are confident enough to have it considering its high F value and tiny p-value.

```
                        GVIF Df GVIF^(1/(2*Df))
log_age_of_exp      1.007965  1        1.003974
log_price           1.778361  1        1.333552
review_scores_value 1.022306  1        1.011091
sqrt_accommodates   1.449189  1        1.203822
host_response_rate  2.242759  1        1.497585
room_type           1.359834  2        1.079870
host_response_time  2.280293  3        1.147269
```

(Check of collinearity)

All predictors exhibited adjusted GVIF values well below 2, suggesting no concerning multicollinearity in the model. This result supports the stability and interpretability of the regression coefficients. Notably, even categorical variables such as room_type and host_response_time, which could potentially inflate multicollinearity due to dummy coding, remained within a safe range. Therefore, we can be confident that the predictors contribute independent and meaningful explanatory power to the model.

model <chr>	adjR2 <dbl>	AIC <dbl>	BIC <dbl>
log_model_with_cat	0.2155836	18620.61	18680.45
log_model_with_resp_time	0.2226245	18572.20	18651.98

Ultimately, we want to consolidate the findings from all models and make the best model selection. Aside from the ANOVA tests we run on each model, AIC, BIC, and adjusted R square will be a combination of criteria we utilize to assess final choice. From the plot we generated, the full model (with two categorical predictors and five continuous predictors) generally outperforms the original one we fit.

In fitting our multiple linear regression model (MLR), we carefully examined each continuous and categorical predictor to ensure the model meets all assumptions of linear regression and is correctly specified. \\

Reiterating what we have previously discussed in section 2 in further detail, we introduced several transformations of the predictors, each of which can be justified based on the data and model requirements. \\

The first transformation we apply is a logarithmic transformation on our response variable total_revenue to reduce its heavy right-skew, as seen in Figure 2.1 and 2.2 before. Although normality of response variable is not assumed in MLR, we still find advantages of doing this. For instance, if the response variable is highly skewed, the residual is likely to be skewed as well, violating an important regression assumption. In such cases, a $\log$ transformation can normalize the distribution of residuals and even improve model fit and inference. Furthermore, after comparing the two scatterplot matrices in Figure 2.3 and 2.4 respectively, we observe an increase in linearity and homoskedasticity between predictor and response variables as a result of our transformation, especially in age_of_exp and price. \\

Moreover, we apply a logarithmic transformation on age_of_exp and price and a square root transformation on accommodates to further improve linearity between these predictors and the response variable, as shown in Figure 2.5. \\

To further explore the potential of transformations beyond what we've already done in section 2, we tested the box-cox transformation on top of our log-transformed response variable. Our expectation is that it can stabilize variance and improve model fit for regression. However, this method yields minimal significant improvements while making interpretation much more difficult. Taking these factors into account, we decided that the transformations in our preliminary model are sufficient.

To ensure that our model fully utilizes the information in our given dataset while maintaining simplicity, we performed an analysis on the performance of our model after potential inclusions and exclusions of predictor variables using partial F-tests. \\

One potential candidate for predictor exclusion was the categorical variable room_type (3 categories), as it was the only one that contributed a predictor variable with a large coefficient p-value of 0.112. However, our partial F-test between our original model ($p+1=8$) and reduced model ($p+1=6$) gives a p-value of 0.000189, which is significantly smaller than 0.05, providing evidence against removing room_type. This leads us to deciding to keep the predictor in our final model.\\

We also have a categorical variable potential candidate to include as a predictor in our final model: The response time of a host (host_response_time). This potential predictor contains 4 categories: $\{\text{within an hour, within a few hours, within a day, a few days or more}\}$. As shown in Figure 4.1, our box-plot indicates a potential pattern with our response variable, but seemed to be minute (this was the primary reason for its exclusion in our preliminary model). However, our partial F-test indicates that adding the new category to our model ($p+1=11$) compared to our original model ($p+1=8$) yields a p-value of $9.605\cdot 10^{-12}$, which is also significantly smaller than 0.05 providing evidence for adding host_response_time. Furthermore, a 10-fold Cross Validation tells us our new model sees an improvement in bias-corrected estimate by 0.01255771; our new model has a VIF of less than 2 for each predictor variable, telling us that multicollinearity is not a problem. This leads us to deciding to add the categorical variable host_response_time as 3 indicator predictors to our model.\\

A quick check of the residual plots of our new model shows that the conclusions reached about our preliminary model regarding assumptions of linear regression holds for our new model as well, as shown in Figure 4.2. Thus, we will finalize our model's predictor variables as the same as our preliminary model and including host_response_time.

To examine the potential existence of problematic observations in our data, we ran an analysis on our finalized model to find outliers and influential points, leading us to decide to keep all observations in our data. \\

For outliers, we use the criterion $|r_i|>4$, where r_i is the standardized residual of the i-th observation, to determine an outlier. This is the standard convention for datasets with more than 50 observations (our dataset contains 5702 observations). As shown in Figure 4.3, all points have a standard residual between 4 and -4, indicating that no outliers are present. This leads to no motivation for finding leverage points as any such points are considered beneficial for the model since they are not outliers.\\

For influential points, we use the Cook’s distance D_{i} as an indicator, with the criterion $$D_{i}>F_{0.5}(11, 5702-11) = 0.9402015$$, where $F_{q}(m,n)$ is the cumulative distribution function for the F-distribution at quantile q with degree of freedom m and n respectively. As observed in Figure 4.3, the Cook's distance of each observation is clearly below our defined threshold, indicating that no highly influential points are present.\\

Combining our results above, we have no outliers nor influential points, indicating no problematic observations in our data. Thus any removal of data will be unjustified.

Final Model Inference and Results

- Create a table that summarizes your final model (coefficients, standard errors, confidence intervals, p-values).

- Provide interpretations of all your coefficients in the context of the problem. Discuss what these estimates reveal about a possible answer to the research question. Be sure to address the specific questions.

- Assess model performance using appropriate metrics (e.g., R-squared, Adjusted R-squared, AIC, BIC) and explain their implications.

```
                                     Estimate Std. Error t value Pr(>|t|)    
(Intercept)                           0.24367    0.29781   0.818  0.41327    
log_age_of_exp                        2.31945    0.07879  29.438  < 2e-16 ***
log_price                             0.49851    0.04534  10.995  < 2e-16 ***
review_scores_value                   0.11014    0.01912   5.760 8.83e-09 ***
sqrt_accommodates                     0.30480    0.04719   6.459 1.14e-10 ***
host_response_rate                    0.74058    0.16573   4.469 8.02e-06 ***
room_typePrivate room                -0.10541    0.04810  -2.191  0.02847 *  
room_typeShared room                 -0.86998    0.21863  -3.979 7.00e-05 ***
host_response_timewithin a day       -0.39293    0.13594  -2.891  0.00386 ** 
host_response_timewithin a few hours -0.25007    0.14884  -1.680  0.09298 .  
host_response_timewithin an hour     -0.08290    0.15310  -0.541  0.58820    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
                                           2.5 %      97.5 %
(Intercept)                          -0.34014877  0.82748392
log_age_of_exp                        2.16498619  2.47391124
log_price                             0.40962303  0.58738998
review_scores_value                   0.07265576  0.14761864
sqrt_accommodates                     0.21228550  0.39731437
host_response_rate                    0.41568995  1.06547573
room_typePrivate room                -0.19970733 -0.01110617
room_typeShared room                 -1.29856825 -0.44139089
host_response_timewithin a day       -0.65942018 -0.12644943
host_response_timewithin a few hours -0.54184085  0.04170980
host_response_timewithin an hour     -0.38302772  0.21723194
```

Our final model is :

【insert final model】

This model provides an estimate of the relations between log of total revenue and 7 predictors. Amongst the 7 predictors, 5–log of age of experience, log of price, review scores value, squareroot of accommodates, host response rate–are continuous, and 2–room type and host response time–are categorical.

The intercept of this model is 0.24367, suggesting that an entire home or apartment with host response time of a few days or more with all continuous variables being 0 has an estimate of expected log of total revenue of 0.24367. However, this intercept lacks practical meanings, as all continuous variables are positive in real life.

The value of all 5 continuous predictors are positively associated with log of total revenue according to their coefficients, meaning that the increase in each continuous predictor result in a release in log of total revenue. An increase of 1 unit in log age of experience, log price, review scores value, square-root of accommodates and host response rate will result in an increase of 2.31945, 0.49851, 0.11014, 0.30480 and 0.74058 in our estimation of average log of total revenue, respectively. All coefficients of the continuous variables have a p-values less than 0.05, suggesting that we reject null hypothesis for each coefficient at 95% significance level. Alternatively, the 95% confidence interval also supports this observation. We are 95% confident that the coefficients fall in such intervals greater than 0, confirming the positive relations between these continuous predictors and log of total revenue.

Room type also affects log of total revenue. The model shows that holding all other predictors constant, changing from renting out an entire home or apartment to a private room causes a decrease of 0.10541 in log of total revenue, and changing to renting out a shared room causes a decrease of 0.86998. Both coefficients have a p-value lof less than 0.05, suggesting that we reject null hypothesis for both coefficients at 95% significant level. We are 95% confidence that the true coefficient for private room falls in [-0.1997, -0.01111], and the true coefficient for shared room falls in [-1.29857, -0.44140], confirming the negative impact on log of total revenue when switching from entire home or apartment to these two room types.

The model also reveals relations between host response time and log of total revenue. Holding all other predictors constant, host response time changing from a few days or more to within a day causes a decrease of -0.39293 in estimated average of log of total revenue. Its p-value of 0.00386 suggests that the coefficient is at 95% significance level, and we are 95% confident that the true coefficient lies between -0.65942 and -0.12645. Changing from a few days or more to a few hours and to within an hour have less significant negative impact. The estimated average of decrease in log of total revenue is 0.25007 when changing to a few hours, and 0.08290 when changing to within an hour. The coefficient of a few hours has a significance level of 90%, while that of within an hour has a significance level of less than 50%. The impact of host response time on log of total revenue is less reliable statistically, especially the response time of within an hour. This is also reflected by the 95% confidence interval–we are 95% confident that the true coefficient of a few hours falls in [-0.54184, 0.04171] and that of less than an hour falls in [-0.38303, 0.21723]. Since 0 in both interval, it is likely that these two response time don’t have a significant impact on log of total revenue. However, since we observed am association between host response time and log of total revenue from the boxplots (Figure x.x), and the estimate of coefficients align with the observation, where host response time causes a relatively

higher estimate in average of log of total revenue, and response time within the day causes the largest decrease, we keep this categorical predictor
to explain more differences in our estimation.

According to the R^2 value, the model explains 22.4% of the differences in log of total revenue. The remaining 77.6% might stem from remaining variables or random noise. Previous diagnostics suggests that no multicollinearity is found between our current predictors, and hence one way to improve this model could be screening through more predictors to explain the differences in our response. However, due to the nature of social science research, an R^2 of 22.4% with most predictors being significant is an acceptable range for this model (Ozili 2023).

model <chr>	adjR2 <dbl>	AIC <dbl>	BIC <dbl>
Preliminary Model	0.215583586	18620.61237	18680.44952
Final Model	0.222624459	18572.19549	18651.97835
Differences	0.007040873	-48.41689	-28.47117

Comparing to our preliminary model, our final model slightly improves in its performance. It improves adjusted 0.007, suggesting that the final model explains slightly more differences in our response, log of total revenue. With a sample size of 5702, we further assess model performance by Akaike Information Criterion (AIC) and Beysian Information Criterion (BIC). A decrease of 48.41689 in AIC suggests that our final model is a better fit than our preliminary model with current complexity level, and a decrease of 28.47117 in BIC suggests that favouring simpler model, our final model with one more predictor still presents a better fit. All these three criteria combined suggest that though very slightly, our final model presents a better fit than the preliminary model. Including the host response time in our final model improves our model performance in general.

Discussion and Conclusion

The final model:

$\log(\text{Total Revenue}) = 0.24367 + 2.31945 * \log(\text{Experience}) + 0.29852 * \log(\text{Price}) + 0.11014 * (\text{Review Score}) + 0.30480 * \sqrt{\text{Accommodate}} + 0.74058 * (\text{Host Response Rate}) - 0.10541 * I(\text{Private Room}) - 0.86998 * I(\text{Shared Room}) -0.39293 * I(\text{Response Time: within a day}) - 0.25007 * I(\text{Response Time: within a few hours}) - 0.08290 * I(\text{Response TimeL within an hour})$

This study aims to identify dominant factors that influence total revenue for Airbnb using multiple linear regression models. The final model incorporated five continuous variables: host business experience, price, customer review scores, accommodation capacity, and host responding speed to be the key predictors. Among all these predictors, year of experience and host response rate seem to be the most influential variables. The coefficient of price reflects the price elasticity of revenue. With an estimated positive coefficient, it suggests that Airbnb revenue is price inelastic and implies that modest price increases may improve earnings (Gunter & Önder, 2018).

Besides, the model involves two categorical variables: room types and host response time. We find that private rooms and the entire home structure do not show a significant difference in the impact on total revenue. In contrast, shared rooms are associated with a substantial reduction in earnings, suggesting guests place a premium on privacy. Also longer the responding time reflects the more negative impact on total revenue.

The final linear regression model presents multiple R^2 equal to 0.224 and adjusted R^2 equal to 0.2226, meaning the model approximately explains 22% of the variance in log-transformed total revenue. The statistical outcome suggests our final model comparably captures meaningful and strong relationships while maintaining the simplicity of the model. Although the figure may not appear to be significant, it is still reasonable in a real-world context, where a large proportion of variation is due to external factors, such as seasonal pattern, regional policy, holiday date, etc (Ozili 2023).

Our finding offers several practical insights for Airbnb hosts and prospective property investors. Firstly, our model identifies both response time and rate as strong predictors of total revenue. The host should respond to customers as quickly and consistently as possible to maintain substantial income. For example, an automatic reply system is encouraged. The hosting experience is another key determinant, meaning the longer a host has been active in doing Airbnb business, the more revenue they tend to generate, as they tend to be more reliable and trustworthy. Therefore, we suggest that new hosts should learn from more experienced business owners and try their best to maintain consistent quality to construct a long-term performance. Moreover, the model suggests that higher review scores tend to increase revenue, despite the influence not being as significant as other factors. The host should prioritize guest experience, offer positive communication, maintain cleanliness, and make sure that all the facilities operate normally. Price strategy is important when making business decisions. The corresponding coefficient is modest at around 0.3, reflecting that revenue will increase with price, but at the same time, the quality of service and room facilities that match the price are of vital importance. Lastly, the host should consider the room type carefully. A shared room tends to generate significantly less income. If feasible, offering private rooms or entire homes may substantially boost earnings.

Although the model is statistically robust and interpretable, several limitations still exist. Our analysis is based on Airbnb listings in the Netherlands, which may limit generalizability to other regions with different tourist dynamics or regulations. Therefore, further improvement could expand the dataset to include multiple countries. Another limitation is that we did not apply automated variable selection techniques during initial model development, such as best subset selection, forward selection, or backward elimination. Instead, we manually selected predictors, and only later used best subset selection to evaluate the final set of predictors. Despite our final model performing well and being supported by statistical metrics, a more rigorous approach would involve automated selection methods at the beginning, which might reduce subjective bias through selecting variables based on data-driven criteria. Automated selection does not necessarily lead to a global optimum, but might be beneficial. Moreover, we did not split the original dataset into training and testing subsets before performing model selection. Ideally, a portion of the data should be reserved for testing, so that the model can be trained on one subset and then validated on unseen data. In future studies, implementing a formal train-test would help assess how well the model generalizes beyond the training sample and guarantee the model's validity. Despite these limitations, our analysis provides a valuable starting point for future Airbnb revenue prediction.

Contributions

- Cruise Chen:
- Qinjue Jiang:
- Fanshi Lin:
- Ray Liu:
- Hanrui Zhang:

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